PPC PHYSICAL VACUUM, RESIDUAL GRAVITATIONAL PRESSURE, AND MERCURY'S PERIHELION PRECESSION

A Three-Case Vacuum Framework and Its Proposed Post-Schwarzschild Correction

Research Paper 1

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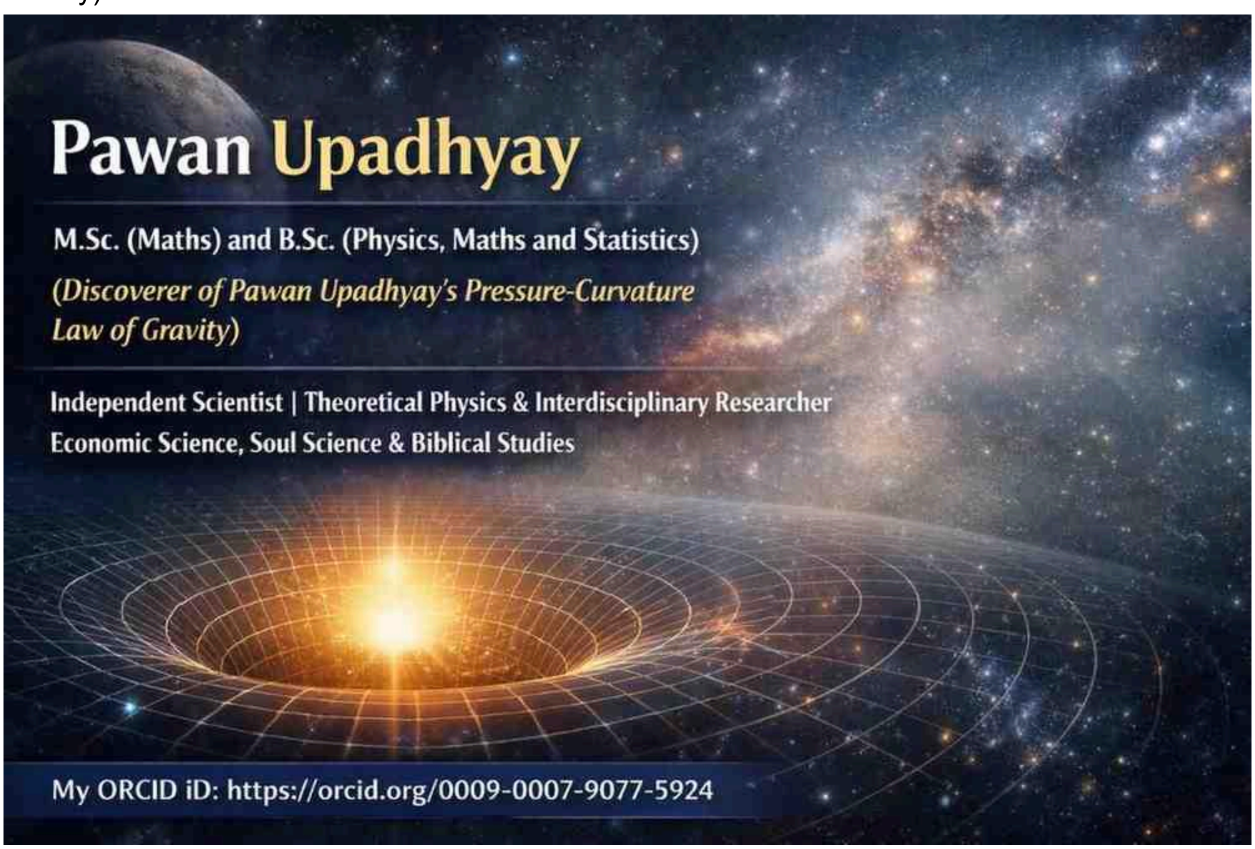
Proposed Theory: Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)



ABSTRACT

The concept of vacuum requires careful distinction between an exact mathematical vacuum, a physical vacuum containing very small residual energy and gravitational-pressure fields, and an asymptotic vacuum in which these quantities approach zero.

This paper develops a three-case framework within the proposed PPC formulation of gravity.

In Case 1, the exact mathematical vacuum is defined by $E_d = 0$, $P_g = 0$, $\nabla P_g = 0$, and $R = 0$. The PPC metric is then identified with the Schwarzschild metric, $\beta_{PPC} = 0$, and the corresponding Mercury perihelion advance is 42.98 arcseconds per century.

Case 2 represents the proposed physical PPC vacuum. In this case, E_d and P_g are very small but finite, while ∇P_g is very small and generally nonzero. The curvature scalar R is also very small and generally nonzero. The resulting PPC metric is therefore proposed to be near-Schwarzschild rather than exactly Schwarzschild. The proposed value $\beta_{PPC} \approx 1.305 \times 10^{-3}$ produces a calculated Mercury perihelion advance of approximately 42.9987 arcseconds per century.

Case 3 describes the asymptotic limit in which E_d, P_g, ∇P_g, R, and β_{PPC} approach zero. In this limit, the PPC metric approaches the Schwarzschild metric and the perihelion prediction approaches 42.98 arcseconds per century.

The three-case framework provides a conceptual distinction between an exact mathematical vacuum and a physical vacuum containing small but finite PPC fields. The Case 2 numerical correction is presented as a proposed PPC result and requires an independent derivation of β_{PPC} from the underlying PPC field equations before it can be regarded as an established prediction.

1. INTRODUCTION

In gravitational theory, the word "vacuum" can represent different mathematical and physical situations.

An exact mathematical vacuum may be defined by the complete absence of the relevant source terms. A physical vacuum, however, may be treated differently within a theory if the theory permits small residual energy-density or pressure fields to remain present even where ordinary matter density is extremely low.

The PPC framework introduces gravitational pressure P_g and matter-energy density E_d as fundamental quantities. This creates the possibility of distinguishing between an exact source-free vacuum and a physical vacuum in which the relevant fields remain very small but finite.

The present paper develops this distinction through three cases:

Case 1 — Exact mathematical vacuum

Case 2 — Physical PPC vacuum

Case 3 — Asymptotic vacuum

The purpose is not to replace the Schwarzschild vacuum solution but to examine how a small residual PPC vacuum structure could produce a geometry that is close to, but not exactly identical to, the Schwarzschild geometry.

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2. PPC VACUUM VARIABLES

The PPC formulation uses the matter-energy density E_d and gravitational pressure P_g as fundamental quantities.

The energy density is represented by:

$$E_d = \rho c^2$$

and the gravitational pressure is represented through the PPC equation of state:

$$P_g = \omega E_d$$

The spatial variation of gravitational pressure is represented by:

$$\nabla P_g$$

Within the PPC framework, the pressure gradient is associated with the gravitational field through the proposed force-density relation:

$$F = \rho g = -\nabla P_g$$

Therefore, even when the energy density and gravitational pressure are very small, a nonzero pressure gradient may remain possible if P_g varies spatially.

This distinction becomes important when considering a physical vacuum rather than an exact mathematical vacuum.

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3. THREE-CASE VACUUM FRAMEWORK

The proposed PPC framework distinguishes the following three cases.

Case 1 — Exact Mathematical Vacuum

The exact vacuum is defined by:

$E_d = 0$

$P_g = 0$

$\nabla P_g = 0$

$R = 0$

In this limiting case, the PPC correction parameter is:

$\beta_{PPC} = 0$

The PPC metric becomes the Schwarzschild metric:

$g_{\mu\nu}^{PPC} = g_{\mu\nu}^{Schw}$.

The corresponding Mercury perihelion advance is:

$\Delta\varphi = 42.98$ arcseconds per century

This case represents the exact mathematical source-free limit.

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4. CASE 2 — PHYSICAL PPC VACUUM

The proposed physical PPC vacuum differs from the exact mathematical vacuum.

In Case 2:

E_d = very small, finite

P_g = very small, finite

∇P_g = very small

and:

R = very small, generally nonzero

Consequently, the PPC metric is proposed to be:

$g_{\mu\nu}^{PPC}$ ≈ near-Schwarzschild

rather than exactly equal to the Schwarzschild metric.

The proposed PPC correction parameter is:

$\beta_{PPC} \approx 1.305 \times 10^{-3}$

and the corresponding proposed Mercury perihelion advance is:

$\Delta\varphi \approx 42.9987$ arcseconds per century

The physical interpretation is that the vacuum is not mathematically empty. Instead, it contains a very small residual PPC energy-density and gravitational-pressure structure.

Because P_g is spatially varying, even a very small value of P_g can produce a small pressure gradient:

$\nabla P_g \neq 0$

This residual gradient is proposed to contribute to the gravitational structure of the vacuum.

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5. RESIDUAL CURVATURE IN THE PPC VACUUM

In Case 1, the curvature quantity R is set to zero.

In Case 2, however:

R = very small, generally nonzero

This represents a fundamental distinction between the two cases.

The physical PPC vacuum is therefore not treated as an exactly flat or exactly source-free mathematical state with respect to the PPC residual fields.

Instead, it is treated as a state that is extremely close to the standard vacuum solution while retaining a small residual gravitational structure.

The proposed geometry can consequently be described as:

PPC physical vacuum → near-Schwarzschild geometry

rather than:

PPC physical vacuum → exactly Schwarzschild geometry

The magnitude of this departure is represented in the proposed framework by β_PPC.

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6. THE PPC CORRECTION PARAMETER

The parameter β_PPC is introduced in the proposed three-case framework as a measure associated with the PPC departure from the exact Schwarzschild vacuum result.

For Case 1:

$\beta_{PPC} = 0$

For Case 2:

$\beta_{PPC} \approx 1.305 \times 10^{-3}$

For Case 3:

$\beta_{PPC} \to 0$

Thus, β_PPC behaves as a correction that disappears in the exact and asymptotic vacuum limits but remains finite in the proposed physical PPC vacuum.

The physical meaning and exact mathematical derivation of β_PPC must be established from the fundamental PPC equations if it is to serve as an independently derived physical parameter.

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7. MERCURY PERIHELION PRECESSION

The three cases lead to three corresponding predictions.

Case 1 — Exact Vacuum

$\beta\_PPC = 0$

$\Delta\varphi = 42.981$ arcseconds per century

This corresponds to the Schwarzschild result used as the reference vacuum prediction.

Case 2 — Physical PPC Vacuum

$\beta\_PPC \approx 1.305 \times 10^{-3}$

$\Delta\varphi\_PPC \approx 42.9987$ arcseconds per century

$\Delta\varphi\_GR = 42.98$

This is the proposed PPC value shown in the present framework.

Case 3 — Asymptotic Vacuum

As the PPC vacuum quantities approach zero:

$E\_d \to 0$

$P\_g \to 0$

$\nabla P\_g \to 0$

$R \to 0$

$\beta\_PPC \to 0$

the metric approaches the Schwarzschild metric:

$g\_\mu\nu PPC \to g\_\mu\nu Schw.$

and the perihelion prediction approaches:

$\Delta\varphi \to 42.981$ arcseconds per century

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8. COMPARATIVE SUMMARY

The three cases can be summarized as follows.

CASE 1 — EXACT VACUUM

$E_d = 0$

$P_g = 0$

$\nabla P_g = 0$

$R = 0$

$g_{\mu\nu}^{PPC} = g_{\mu\nu}^{Schw}$.

$\beta_{PPC} = 0$

$\Delta\varphi = 42.981''$

CASE 2 — PHYSICAL PPC VACUUM

E_d = very small, finite

P_g = very small, finite

∇P_g = very small

R = very small, generally nonzero

$g_{\mu\nu}^{PPC}$ = near-Schwarzschild

$\beta_{PPC} \approx 1.305 \times 10^{-3}$

$\Delta\varphi \approx 42.9987''$ (proposed)

CASE 3 — ASYMPTOTIC VACUUM

$E_d \to 0$

$P_g \to 0$

$\nabla P_g \to 0$

$R \to 0$

$g\_{\mu\nu PPC} \to g\_{\mu\nu Schw.}$

$\beta\_{PPC} \to 0$

$\Delta\varphi \to 42.981''$

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9. PHYSICAL INTERPRETATION

The central proposition of this framework is that the word "vacuum" need not always be identified with an exactly zero value of every physical field.

The framework distinguishes:

Exact mathematical vacuum

from

physical PPC vacuum

and from

asymptotic vacuum.

In the physical PPC vacuum, the quantities E_d and P_g are proposed to remain very small but finite.

Consequently, their spatial variation may produce a very small pressure gradient:

$\nabla P\_g \neq 0$

although its magnitude is very small.

This provides a possible PPC mechanism by which a vacuum region can remain extremely close to the Schwarzschild solution while producing a small departure from the exact Schwarzschild prediction.

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10. RELATION TO THE SCHWARZSCHILD LIMIT

The framework does not reject the Schwarzschild solution.

Instead, the Schwarzschild geometry appears naturally as a limiting case.

The proposed relationship is:

Physical PPC vacuum

↓

very small E_d and P_g

↓

very small ∇P_g and R

↓

near-Schwarzschild metric

↓

small PPC correction

↓

$\beta_{PPC} \neq 0$

As the residual PPC fields disappear:

$E_d, P_g, \nabla P_g, R \to 0$

the correction disappears:

$\beta_{PPC} \to 0$

and the standard Schwarzschild result is recovered:

$g_{\mu\nu}^{PPC} \to g_{\mu\nu}^{Schw}$.

$\Delta\varphi \to 42.981$ arcseconds per century

11. SCIENTIFIC STATUS OF THE PROPOSED CORRECTION

The distinction between the three vacuum cases is conceptually clear within the proposed PPC framework.

However, the numerical value:

$\beta\_PPC \approx 1.305 \times 10^{-3}$

and the resulting:

$\Delta\varphi \approx 42.9987$ arcseconds per century

should be identified as a **proposed PPC result** unless the underlying PPC field equations independently derive β_PPC and the corresponding orbital correction.

In particular, the theory should ultimately demonstrate:

1. How E_d and P_g remain finite in the physical vacuum.
2. The exact spatial dependence of P_g.
3. How ∇P_g is calculated.
4. How the residual fields determine R.
5. How the PPC fields modify the metric.
6. How β_PPC follows mathematically from those fields.
7. How the modified metric produces the stated perihelion correction.

This derivation would distinguish an independently derived PPC prediction from a parameter introduced to reproduce an observed or desired correction.

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12. CONCLUSION

This paper establishes a three-case conceptual framework for distinguishing different meanings of vacuum within the proposed PPC gravitational formulation.

Case 1 represents the exact mathematical vacuum, where E_d, P_g, ∇P_g, and R are zero. The PPC metric is Schwarzschild and β_{PPC} is zero.

Case 2 represents the proposed physical PPC vacuum, in which E_d and P_g are very small but finite, ∇P_g is very small, and R is very small and generally nonzero. The resulting metric is proposed to be near-Schwarzschild. The framework assigns:

$\boldsymbol{\beta_{PPC} \approx 1.305 \times 10^{-3}}$

and proposes:

$\Delta\varphi \approx 42.9987$ arcseconds per century

for Mercury.

Case 3 represents the asymptotic limit in which all residual PPC quantities approach zero. In this limit:

$g_{\mu\nu PPC} \to g_{\mu\nu Schw}$.

$\beta_{PPC} \to 0$

and:

$\Delta\varphi \to 42.981$ arcseconds per century.

The framework therefore provides a distinction between an exact mathematical vacuum and a physical PPC vacuum containing very small residual energy density, gravitational pressure, pressure gradient, and curvature.

The principal task for further development is to derive the Case 2 correction directly from the fundamental PPC field equations rather than treating β_{PPC} as an independently assumed parameter.

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SOURCE AND SCOPE NOTE

The three-case values and terminology in this paper are based on the supplied PPC framework shown in the accompanying table. The Case 2 value $\beta_{PPC} \approx 1.305 \times 10^{-3}$ and the corresponding 42.9987″ perihelion value are therefore presented as **proposed PPC results**, not as independently verified results established by this paper.

The purpose of this paper is to document and clarify the proposed three-case vacuum framework and its relationship to the Schwarzschild limiting case.

SPECIAL NOTE

VACUUM AND REAL LABORATORY VACUA

1. Meaning of Vacuum

In theoretical physics, the term **vacuum** can have a precise mathematical meaning. In a classical gravitational calculation, a vacuum region is commonly treated as a region in which the relevant matter-energy source terms are absent.

For an ideal source-free vacuum:

$$E_d = 0$$

$$P_g = 0$$

$$\nabla P_g = 0$$

Under these assumptions, the PPC source tensor vanishes:

$$T_{\mu\nu} = 0$$

and the field equation becomes:

$$G_{\mu\nu} = 0$$

This represents the exact mathematical vacuum limit used when deriving the Schwarzschild exterior.

However, the word **vacuum** does not necessarily mean that a physically realizable region contains absolutely nothing.

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2. Real Laboratory Vacuum

A **real laboratory vacuum** is a physical region from which most of the ordinary matter, particularly gas molecules, has been removed using vacuum equipment.

Examples include vacuum chambers, vacuum tubes, experimental apparatus, and particle-physics vacuum systems.

A laboratory vacuum is therefore a **low-density physical environment**, not necessarily an absolutely empty region.

Even after extensive pumping, a real laboratory vacuum can contain extremely small residual contributions, including:

- residual gas molecules;
- particles released from surrounding surfaces;
- electromagnetic radiation;
- thermal radiation;
- other physical fields or background contributions.

Consequently, for a real laboratory vacuum, it is generally more appropriate to distinguish between **exact zero** and **extremely small values**.

Instead of automatically writing:

$$E_d = 0$$

one may describe the physical situation schematically as:

$$E_d \approx 0$$

Similarly:

$P_g \approx 0$

and, where a spatial variation exists:

$\nabla P_g \approx 0$

These quantities can be extremely small while still being finite.

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3. Exact Vacuum Versus Real Laboratory Vacuum

The distinction can therefore be summarized as follows.

Ideal mathematical vacuum

$E_d = 0$

$P_g = 0$

$\nabla P_g = 0$

The quantities are defined to vanish exactly.

Real laboratory vacuum

$E_d \approx 0$

$P_g \approx 0$

$\nabla P_g \approx 0$

The quantities are extremely small and may be treated as zero only when the required experimental or theoretical precision permits that approximation.

Thus, **"vacuum" and "absolutely empty space" should not automatically be treated as synonymous**.

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4. Relevance to the PPC Framework

This distinction is particularly relevant to the PPC formulation because PPC assigns physical significance to the energy density E_d, gravitational pressure P_g, and pressure gradient ∇P_g.

The proposed PPC physical-vacuum description can therefore distinguish between:

Exact mathematical vacuum

and

Physical vacuum with very small residual fields.

In the physical-vacuum case, the proposed framework may have:

E_d = very small, finite

P_g = very small, finite

∇P_g = very small

rather than requiring all three quantities to be exactly zero.

This is consistent with the three-case framework in which the physical PPC vacuum is represented separately from the exact mathematical vacuum.

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5. Important Scientific Distinction

The existence of a real laboratory vacuum does **not by itself prove** that a particular residual energy density or gravitational-pressure field exists.

A statement such as:

$E_d > 0$

or

$P_g > 0$

in a PPC physical vacuum is a **theoretical assumption or prediction of the PPC framework** unless it is independently derived or experimentally established.

Therefore, the following concepts should be kept separate:

Experimental fact:
A laboratory vacuum contains extremely little ordinary matter.

Theoretical description:
A physical vacuum may be assigned residual energy or field quantities.

PPC hypothesis:
A very small finite E_d, P_g, and ∇P_g may remain in the physical vacuum and may contribute to gravitational behavior.

This distinction prevents the term **“real laboratory vacuum”** from being used as direct evidence for a specific PPC prediction without an independent measurement or derivation.

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6. Relation to the Three-Case PPC Framework

The three cases can therefore be interpreted as:

Case 1 — Exact mathematical vacuum

$E_d = 0$
$P_g = 0$
$\nabla P_g = 0$

Case 2 — Physical PPC vacuum

E_d = very small, finite
P_g = very small, finite
∇P_g = very small

Case 3 — Asymptotic vacuum

$E_d \to 0$
$P_g \to 0$
$\nabla P_g \to 0$

Case 1 is an exact mathematical limit.

Case 2 represents the proposed physical PPC vacuum.

Case 3 represents the limiting behavior as the residual quantities become progressively smaller and approach zero.

This distinction allows the PPC framework to discuss a physical vacuum without conflating it with the exact source-free mathematical vacuum used in the Schwarzschild solution.

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7. Conclusion

A **vacuum** in theoretical physics can be an ideal mathematical condition in which specified source quantities are exactly zero.

A **real laboratory vacuum** is instead a physical low-density environment in which ordinary matter has been greatly reduced but is not necessarily eliminated completely.

Therefore:

Ideal vacuum → exact zero in the mathematical model

Real laboratory vacuum → extremely small residual physical contributions

Within the PPC framework, this distinction permits the proposed physical vacuum to contain very small finite values of E_d and P_g and a very small pressure gradient ∇P_g, while the exact mathematical vacuum remains the limiting case in which these quantities are zero.

The broader PPC treatment of vacuum energy density, gravitational pressure, and related fields should be considered together with the associated PPC research papers referenced by the author.

SPECIAL NOTE

PHYSICAL VACUUM AND MERCURY'S PERIHELION PRECESSION

Observations establish that Mercury's orbit undergoes a measurable perihelion precession. After accounting for the major Newtonian planetary perturbations and other known contributions, the relativistic contribution is approximately:

42.98 arcseconds per century.

The PPC framework distinguishes this observational result from the idealized mathematical concept of an exact vacuum.

An **exact mathematical vacuum** is represented by:

$$E_d = 0$$
$$P_g = 0$$
$$\nabla P_g = 0$$

However, the PPC framework proposes that the physical near-vacuum through which Mercury moves may not be an exact mathematical vacuum. Instead, very small residual quantities may be present:

$E_d > 0$, but very small

$P_g > 0$, but very small

$|\nabla P_g| > 0$, but very small

Thus, the PPC physical-vacuum concept can be expressed as:

Physical near-vacuum ≠ exact mathematical vacuum

and:

Physical near-vacuum → very small E_d + very small P_g + very small $|\nabla P_g|$

These residual PPC quantities could produce a very small departure from the exact Schwarzschild result.

Observational and theoretical distinction

The observation directly establishes the **orbital phenomenon—**Mercury's measured perihelion precession.

The presence of small **E_d, P_g, and ∇P_g** is the **PPC theoretical interpretation** of the physical environment and its possible contribution to the observed precession.

Therefore, the appropriate scientific statement is:

Mercury's observed perihelion precession demonstrates a measurable relativistic orbital effect. Within the PPC framework, this result is interpreted as being potentially associated with a physical near-vacuum containing very small residual energy density, gravitational pressure, and pressure gradient, rather than an exactly source-free mathematical vacuum.

This interpretation corresponds to the PPC **very small but finite source** case, in which E_d is nonzero but extremely small and $P_g = \omega E_d$ is correspondingly small.

Scientific qualification

The observational value of approximately **42.98 arcseconds per century does not, by itself, directly measure E_d, P_g, or ∇P_g**.

Therefore, the PPC framework must establish quantitatively that the proposed residual quantities produce the corresponding modification of the metric and Mercury's geodesic.

The theoretical chain to be demonstrated is:

E_d, P_g, ∇P_g

→ PPC metric $g_{\mu\nu}$

→ Christoffel symbols Γ

→ Mercury geodesic

→ perihelion correction

→ observed orbital precession

This distinction preserves the empirical observation while identifying the small residual energy density and gravitational pressure as the **PPC physical interpretation to be tested by the theory**.

PPC PHYSICAL VACUUM, RESIDUAL GRAVITATIONAL PRESSURE, AND MERCURY'S PERIHELION PRECESSION

A Three-Case Vacuum Framework and Its Relation to the Schwarzschild Limit

Research Paper 2

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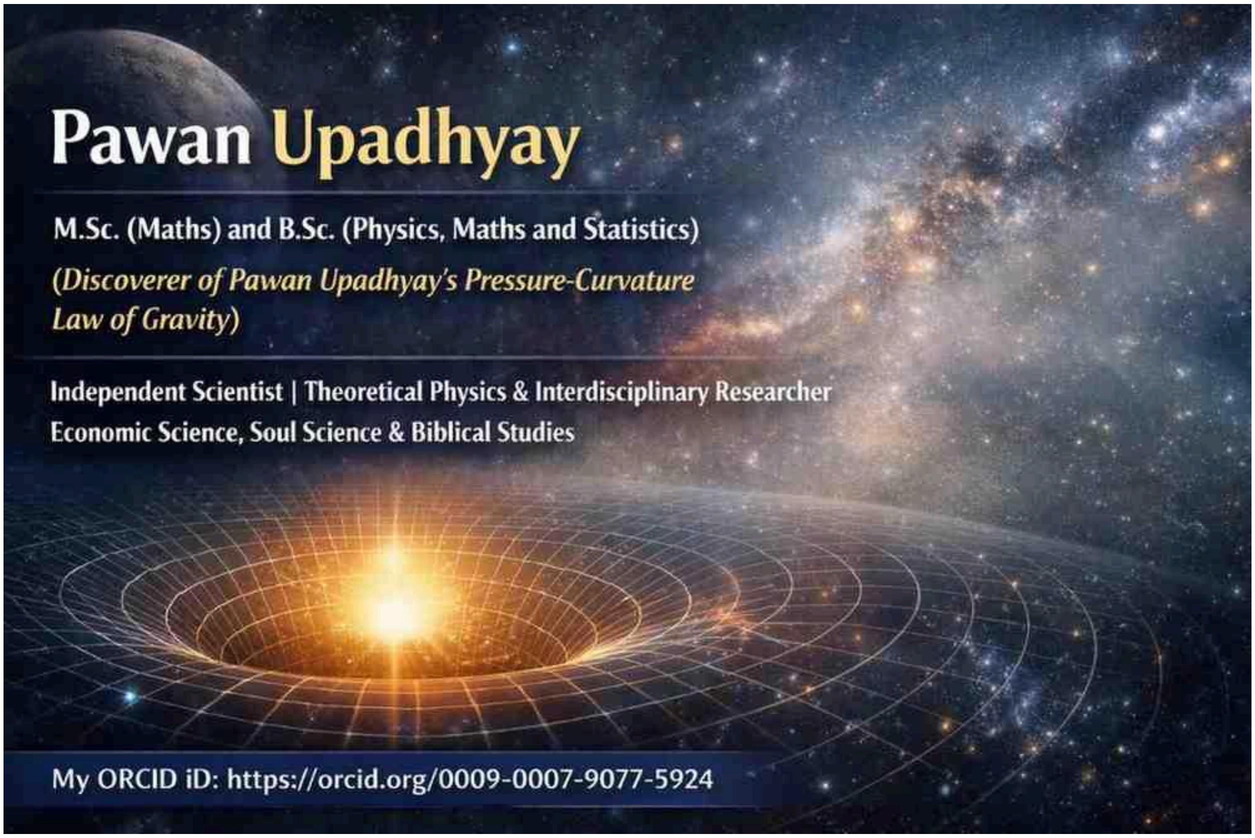
Proposed Theory: Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)



ABSTRACT

The concept of vacuum requires a distinction between an exact mathematical vacuum and a physical vacuum in which extremely small residual energy-density and gravitational-pressure fields may be present.

Within the Pressure Curvature Law of Gravity (PPC) framework, this paper presents three vacuum cases.

Case 1 — Exact Vacuum:
$E_d = 0$, $P_g = 0$, and $\nabla P_g = 0$. The PPC tensor equation reduces to $G_{\mu\nu} = 0$, giving the Schwarzschild exterior and the standard General Relativity contribution to Mercury's perihelion advance, approximately 42.981 arcseconds per century.

Case 2 — Very Small but Finite PPC Source:
E_d is very small but finite, $P_g = \omega E_d$ is very small but nonzero, and the pressure gradient is very small. The PPC metric is therefore expected to differ only slightly from the Schwarzschild metric. The proposed PPC perihelion value is 42.9987 arcseconds per century, corresponding to $\beta_{PPC} \approx 1.305 \times 10^{-3}$.

Case 3 — Continuous Near-Vacuum Limit:
$E_d \to 0$, $P_g \to 0$, and $|\nabla P_g| \to 0$. Provided the PPC field equations have a regular continuous vacuum limit, the PPC metric approaches the Schwarzschild metric, β_{PPC} approaches zero, and the perihelion advance approaches 42.981 arcseconds per century.

The three-case framework separates the exact GR-recovery limit from the proposed finite PPC correction and its asymptotic behavior.

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1. INTRODUCTION

The PPC formulation describes gravitational effects through energy density and gravitational pressure. The foundational PPC equations define:

$$E_d = \rho c^2$$

and

$$P_g = \omega E_d$$

where E_d is the energy-density field, ρ is the matter mass-density field, P_g is the gravitational pressure, and ω is the PPC equation-of-state parameter. The PPC weak-field formulation further associates the combination $E_d + 3P_g$ with the strength of curvature and the pressure gradient ∇P_g with the geometry of curvature and consequently with the path or orbit followed by matter.

The present paper focuses specifically on the meaning of vacuum within this framework.

The central distinction is between:

Exact mathematical vacuum

Very small but finite PPC source

and

Continuous near-vacuum limit.

This distinction allows the exact Schwarzschild recovery and the proposed PPC correction to be considered separately.

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2. PPC ENERGY DENSITY AND GRAVITATIONAL PRESSURE

The PPC formulation defines the energy-density field as:

$$E_d(x, y, z, ct) = \rho(x, y, z, ct)c^2$$

The gravitational pressure is:

$$P_g(x, y, z, ct) = \omega(x, y, z, ct)E_d(x, y, z, ct)$$

Thus, when E_d is exactly zero, the PPC equation of state gives $P_g = 0$.

The PPC formulation also identifies the pressure gradient as an important quantity:

$$\nabla P_g$$

In the PPC interpretation, $E_d + 3P_g$ contributes to the strength of curvature, while ∇P_g is associated with the geometry of curvature and the path followed by matter.

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3. PPC TENSOR EQUATION

The PPC perfect-fluid stress-energy tensor is written as:

$$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

with the gravitational field equation:

$$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$$

Using:

$$P_g = \omega E_d$$

and:

$$E_d = \rho c^2$$

gives:

$$G_{\mu\nu} = (8\pi G/c^2)\rho[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$$

This provides the tensor basis for considering the three vacuum cases.

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4. CASE 1 — EXACT MATHEMATICAL VACUUM

In the exact vacuum limit:

$E_d = 0$

$P_g = 0$

$\nabla P_g = 0$

Therefore:

$T_{\mu\nu} = 0$

and:

$G_{\mu\nu} = 0$

This is the PPC vacuum limit used in the Claim A analysis.

For a static, spherically symmetric exterior, the corresponding solution is the Schwarzschild geometry.

The Mercury perihelion calculation therefore gives the standard General Relativity contribution:

$\Delta\varphi_{GR} \approx 42.981$ arcseconds per century

or approximately:

43 arcseconds per century.

This constitutes the PPC recovery of the GR vacuum prediction.

The significance of this case is that the PPC tensor formulation reproduces the established GR vacuum result when its PPC source terms are set exactly to zero.

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5. CASE 2 — VERY SMALL BUT FINITE PPC SOURCE

The second case considers a regime in which the PPC source quantities are not exactly zero but are extremely small.

The assumptions are:

$E_d > 0$, but $E_d \ll 1$

$\omega > 0$, with $\omega \ll 1$

and therefore:

$P_g = \omega E_d$

is very small but not identically zero.

The pressure gradient may also be very small:

$|\nabla P_g| \approx$ very small

The resulting PPC metric is expected to differ only slightly from the Schwarzschild metric.

This is therefore a **near-Schwarzschild PPC regime**, rather than the exact mathematical vacuum.

The proposed PPC perihelion value is:

$\Delta\varphi_{PPC} = 42.9987$ arcseconds per century

Using the relation:

$\Delta\varphi_{PPC} = \Delta\varphi_{GR}(1 + \beta_{PPC}/3)$

with:

$\Delta\varphi_{GR} = 42.98$ arcseconds per century

gives:

$\beta_{PPC} \approx 1.305 \times 10^{-3}$

This represents the proposed finite-source PPC correction. The underlying PPC metric and Mercury geodesic provide the required mathematical route for deriving this correction.

6. PRESSURE GRADIENT IN THE NEAR-VACUUM REGIME

The PPC pressure gradient is:

$$\nabla P_g = c^2 \nabla(\omega\rho)$$

A small gravitational pressure does not, by itself, mathematically guarantee a small pressure gradient.

Therefore, the near-vacuum condition must separately consider:

$$P_g \approx 0$$

and:

$$|\nabla P_g| \approx 0$$

When both are very small, the PPC pressure-force contribution:

$$\rho g = -\nabla P_g$$

is also very small.

This distinction is important because the magnitude of P_g and its spatial variation are separate mathematical quantities.

7. CASE 2 AND THE PPC METRIC

The PPC weak-field equation is expressed as:

$$\nabla^2\Phi_{PPC} = (4\pi G/c^2)(E_d + 3P_g)$$

In the very weak PPC-source regime:

$$E_d \approx 0$$

and:

$P_g \approx 0$

so the residual PPC source contribution becomes very small.

The weak-field metric approaches the vacuum metric as the PPC source quantities become small.

Thus:

$g_{\mu\nu}PPC \approx$ near-Schwarzschild

and, in the continuous limit:

$g_{\mu\nu}PPC \rightarrow g_{\mu\nu}Schwarzschild$

provided the complete PPC field equations possess a regular continuous vacuum limit.

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8. CASE 3 — CONTINUOUS NEAR-VACUUM LIMIT

The third case considers the limiting process:

$E_d \rightarrow 0$

$P_g \rightarrow 0$

$|\nabla P_g| \rightarrow 0$

Provided the PPC field equations have a regular continuous vacuum limit:

$g_{\mu\nu}PPC \rightarrow g_{\mu\nu}Schwarzschild$

and therefore:

$\beta_{PPC} \rightarrow 0$

Consequently:

$\Delta\varphi_{PPC} \rightarrow$ 42.981 arcseconds per century

This case establishes the continuity of the proposed PPC description with the exact Schwarzschild limit.

9. THREE-CASE SUMMARY

CASE 1 — EXACT VACUUM

$E_d = 0$

$P_g = 0$

$\nabla P_g = 0$

$R = 0$

$g_{\mu\nu}^{PPC} = g_{\mu\nu}^{Schwarzschild}$

$\beta_{PPC} = 0$

$\Delta\varphi = 42.981''$/century

CASE 2 — VERY SMALL BUT FINITE PPC SOURCE

$E_d > 0$, but very small

$P_g = \omega E_d$

P_g = very small

$|\nabla P_g|$ = very small

R = very small, generally nonzero

$g_{\mu\nu}^{PPC} \approx$ near-Schwarzschild

$\beta_{PPC} \approx 1.305 \times 10^{-3}$

$\Delta\varphi_{PPC} = 42.9987''$/century — proposed

CASE 3 — CONTINUOUS NEAR-VACUUM LIMIT

$E_d \to 0$

$P_g \to 0$

$|\nabla P_g| \to 0$

$R \to 0$

$g_{\mu\nu}^{PPC} \to g_{\mu\nu}^{Schwarzschild}$

$\beta_{PPC} \to 0$

$\Delta\varphi_{PPC} \to 42.981''/\text{century}$

---

10. RELATION TO THE PREVIOUS PPC CLAIM A PAPER

The present three-case analysis extends, rather than replaces, the earlier Claim A research paper.

The earlier Claim A paper was deliberately restricted to the exact vacuum case. Its conclusion was:

PPC vacuum tensor equation

$\to G_{\mu\nu} = 0$

→ Schwarzschild exterior

→ relativistic perihelion advance

and it explicitly stated that this establishes recovery of the GR vacuum prediction rather than an independent PPC mechanism.

The present paper adds the finite-source and limiting cases.

Therefore:

Claim A = exact-vacuum GR recovery

Case 2 = proposed finite PPC correction

Case 3 = continuous recovery of the GR limit

These are complementary parts of the PPC research program.

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11. REAL LABORATORY VACUUM AND THE PPC PHYSICAL VACUUM

A real laboratory vacuum should not automatically be identified with the exact mathematical vacuum.

A laboratory vacuum is a physical low-density environment in which most ordinary matter has been removed. It can contain residual matter and other physical contributions.

Consequently, the physical condition may be represented schematically by:

$E_d \approx 0$

rather than necessarily:

$E_d = 0$

Within the PPC framework, if the physical vacuum is proposed to contain a very small residual PPC energy-density and gravitational-pressure structure, it can be represented by:

E_d = very small, finite

P_g = very small, finite

$|\nabla P_g|$ = very small

This should be understood as a **PPC theoretical description**, not as a conclusion that every laboratory vacuum necessarily possesses a measurable PPC gravitational-pressure field.

The exact relationship between laboratory vacuum conditions and the PPC physical-vacuum regime would require independent theoretical derivation and experimental investigation.

---

12. SCIENTIFIC STATUS OF THE THREE CASES

The three cases have different mathematical roles.

Case 1 is the exact source-free limit and provides the GR-recovery result.

Case 2 is the proposed finite PPC-source regime and provides the proposed correction:

$\beta\_PPC \approx 1.305 \times 10^{-3}$

and:

$\Delta\varphi\_PPC = 42.9987''$/century.

Case 3 describes the limiting behavior as the PPC source quantities approach zero.

The 42.9987″ value is specifically a **proposed PPC Case 2 prediction**, not an observational value supplied by NASA. The decisive theoretical test is the derivation:

$E\_d, P\_g \rightarrow g\_\mu\nu PPC \rightarrow \Gamma \rightarrow$ Mercury geodesic $\rightarrow \beta\_PPC \rightarrow \Delta\varphi\_PPC$

as described in the PPC research material.

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13. CONCLUSION

The PPC framework provides a three-case description of vacuum behavior.

The first case is the exact mathematical vacuum:

$E\_d = P\_g = \nabla P\_g = 0$

which gives:

$G\_\mu\nu = 0$

and therefore the Schwarzschild exterior and the standard relativistic Mercury perihelion contribution of approximately:

42.981 arcseconds per century.

The second case considers a very small but finite PPC source. In this regime, E_d and P_g remain extremely small, while the pressure gradient is also very small. The geometry is expected to remain close to Schwarzschild. The proposed PPC correction is:

$\beta\_PPC \approx 1.305 \times 10^{-3}$

with a proposed perihelion prediction of:

42.9987 arcseconds per century.

The third case describes the continuous approach to the exact vacuum:

E_d → 0

P_g → 0

|∇P_g| → 0

so that:

g_μνPPC → g_μνSchwarzschild

β_PPC → 0

and:

Δφ_PPC → 42.981″/century.

The three cases therefore provide a unified framework connecting the exact GR-recovery limit, the proposed finite PPC correction, and the asymptotic Schwarzschild limit.

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[4] Einstein, A. *Die Feldgleichungen der Gravitation*, 1915.

[5] Einstein, A. *Erklärung der Perihelbewegung des Merkur aus der allgemeinen Relativitätstheorie*, 1915.

FINAL NOTE

This paper is intended as a **separate PPC research paper on the physical and mathematical meaning of vacuum**, complementing the foundational PPC equations paper and the Claim A Mercury perihelion paper. It preserves the distinction between the exact vacuum result and the proposed finite PPC-source correction rather than combining them into a single vacuum assumption.

SPECIAL NOTE

CALCULATION OF βPPC USING DIFFERENT GR REFERENCE VALUES

The calculated value of the PPC correction parameter β_{PPC} depends on the reference value used for the General Relativity perihelion advance.

The relation is:

$$\Delta\varphi_{PPC} = \Delta\varphi_{GR} [1 + \beta_{PPC}/3]$$

Therefore:

$$\beta_{PPC} = 3[(\Delta\varphi_{PPC} / \Delta\varphi_{GR}) - 1]$$

For the proposed PPC value:

$$\Delta\varphi_{PPC} = 42.9987 \text{ arcseconds/century}$$

two commonly used GR reference values give slightly different results.

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1. Using the rounded GR value: 42.98″/century

$\beta_{PPC} = 3[(42.9987 / 42.98) - 1]$

42.9987 / 42.98 = **1.0004350861**

Therefore:

$\beta_{PPC} = 3 \times 0.0004350861$

$$\beta_{PPC} = 0.0013052583$$

or:

β_PPC = 1.3052583 × 10^{-3}

Rounded:

β_PPC ≈ 1.305 × 10^{-3}

---

2. Using the more precise GR value: 42.981″/century

β_PPC = 3[(42.9987 / 42.981) − 1]

42.9987 / 42.981 = **1.0004118099**

Therefore:

β_PPC = 3 × 0.0004118099

β_PPC = 0.0012354296

or:

β_PPC = 1.2354296 × 10^{-3}

Rounded:

β_PPC ≈ 1.23543 × 10^{-3}

---

3. Comparison

GR reference value	PPC value	Calculated βPPC
42.98″/century	42.9987″/century	**1.3052583 × 10^{-3}**
42.981″/century	42.9987″/century	**1.2354296 × 10^{-3}**

Thus:

$42.98'' \rightarrow \beta PPC \approx 1.305 \times 10^{-3}$

whereas:

$42.981'' \rightarrow \beta PPC \approx 1.23543 \times 10^{-3}$

---

4. Explanation of the Difference

The difference occurs because βPPC is calculated from the **ratio between the PPC value and the GR reference value**.

When the reference changes from:

$42.98 \rightarrow 42.981$

the ratio changes, and consequently βPPC changes.

Therefore, the value:

$\beta_{PPC} = 1.305 \times 10^{-3}$

is mathematically consistent with:

$\Delta\varphi_{GR} = 42.98''$/century

and:

$\Delta\varphi_{PPC} = 42.9987''$/century.

It is **not** the value obtained when 42.981″/century is used as the reference.

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5. Recommended Reporting

To avoid ambiguity, the research paper should explicitly state the reference value used.

If the intended calculation is based on the rounded GR benchmark:

Using the GR reference value $\Delta\varphi_{GR} = 42.98''$/century and the proposed PPC value $\Delta\varphi_{PPC} = 42.9987''$/century, the PPC correction parameter is $\beta_{PPC} = 1.3052583 \times 10^{-3}$, approximately 1.305×10^{-3}.

If the more precise reference value is used:

Using $\Delta\varphi\_GR$ = 42.981″/century, the corresponding correction parameter is β_PPC = 1.2354296 × 10^{-3}, approximately 1.23543 × 10^{-3}.

Therefore, both calculations are arithmetically correct; they correspond to different choices of the GR reference value.

Additional Images:

Detailed Calculation Note: Determination of the PPC Correction Parameter β_{PPC}

The proposed PPC perihelion relation used in the present framework is

$$\Delta\phi_{\mathrm{PPC}} = \Delta\phi_{\mathrm{GR}}\left(1 + \frac{\beta_{\mathrm{PPC}}}{3}\right).$$

Solving this equation for β_{PPC}:

$$\frac{\Delta\phi_{\mathrm{PPC}}}{\Delta\phi_{\mathrm{GR}}} = 1 + \frac{\beta_{\mathrm{PPC}}}{3},$$

therefore,

$$\frac{\beta_{\mathrm{PPC}}}{3} = \frac{\Delta\phi_{\mathrm{PPC}}}{\Delta\phi_{\mathrm{GR}}} - 1,$$

and hence

$$\beta_{\mathrm{PPC}} = 3\left(\frac{\Delta\phi_{\mathrm{PPC}}}{\Delta\phi_{\mathrm{GR}}} - 1\right)$$

This is the calculation formula stated in the paper.

PPC PHYSICAL VACUUM, RESIDUAL GRAVITATIONAL PRESSURE, AND MERCURY'S PERIHELION PRECESSION.pdf

1. Proposed PPC perihelion value

The proposed PPC value used in the calculation is

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''/\mathrm{century}$$

This is presented in the paper as the proposed Case 2 PPC result.

PPC PHYSICAL VACUUM, RESIDUAL GRAVITATIONAL PRESSURE, AND MERCURY'S PERIHELION PRECESSION.pdf

Calculation A — Using the rounded GR value $42.98''$

Take

$$\Delta\phi_{\mathrm{GR}} = 42.98''/\mathrm{century}.$$

Substituting into the equation:

$$\beta_{\mathrm{PPC}} = 3\left(\frac{42.9987}{42.98} - 1\right).$$

Step 1: Calculate the ratio

$$\frac{42.9987}{42.98} = 1.0004350861.$$

Therefore,

$$\frac{\Delta\phi_{\mathrm{PPC}}}{\Delta\phi_{\mathrm{GR}}} - 1 = 1.0004350861 - 1$$

$$= 0.0004350861.$$

Step 2: Multiply by 3

$$\beta_{\mathrm{PPC}} = 3(0.0004350861)$$

$$= 0.0013052583.$$

Therefore,

$$\beta_{\mathrm{PPC}} = 1.3052583 \times 10^{-3}$$

and, to four significant figures,

$$\beta_{\mathrm{PPC}} \approx 1.305 \times 10^{-3} \ .$$

This reproduces the calculation already contained in the paper.

Calculation B — Using the more precise GR value

$42.981''$

Now take

$$\Delta\phi_{\text{GR}} = 42.981''/\text{century}.$$

The same equation gives

$$\beta_{\text{PPC}} = 3\left(\frac{42.9987}{42.981} - 1\right).$$

Step 1: Calculate the ratio

$$\frac{42.9987}{42.981} = 1.0004118099.$$

Therefore,

$$1.0004118099 - 1 = 0.0004118099.$$

3. Direct comparison

The two calculations can therefore be written as:

$$42.98'' \Rightarrow \beta_{\mathrm{PPC}} = 1.3052583 \times 10^{-3}$$

while

$$42.981'' \Rightarrow \beta_{\mathrm{PPC}} = 1.2354296 \times 10^{-3} \, .$$

The difference is

$$1.3052583 \times 10^{-3} - 1.2354296 \times 10^{-3}$$

$$= 0.0698287 \times 10^{-3}$$

or

$$6.98287 \times 10^{-5} \, .$$

The difference is

$$1.3052583 \times 10^{-3} - 1.2354296 \times 10^{-3}$$

$$= 0.0698287 \times 10^{-3}$$

or

$$6.98287 \times 10^{-5} \ .$$

Thus the numerical value of β_{PPC} changes because the denominator—the selected GR reference value—changes. The paper explicitly identifies this reference-value dependence.

4. Verification by substituting β_{PPC} back

The calculation can also be checked in the reverse direction.

Using

$$\beta_{\mathrm{PPC}} = 1.3052583 \times 10^{-3}$$

and

$$\Delta\phi_{\mathrm{GR}} = 42.98'',$$

we have

$$\frac{\beta_{\mathrm{PPC}}}{3} = \frac{1.3052583 \times 10^{-3}}{3}$$

$$= 0.0004350861.$$

Therefore,

$$1 + \frac{\beta_{\mathrm{PPC}}}{3} = 1.0004350861.$$

Hence,

$$\Delta\phi_{\mathrm{PPC}} = 42.98(1.0004350861).$$

This gives approximately

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''/\mathrm{century}\ .$$

Thus the 1.3052583×10^{-3} value is mathematically consistent with the pair

$$\Delta\phi_{\mathrm{GR}} = 42.98''$$

and

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''\ .$$

5. Verification using $42.981''$

Similarly,

$$\beta_{\mathrm{PPC}} = 1.2354296 \times 10^{-3}.$$

Then

$$\frac{\beta_{\mathrm{PPC}}}{3} = 0.0004118099.$$

Therefore,

$$1 + \frac{\beta_{\mathrm{PPC}}}{3} = 1.0004118099.$$

Substitution gives

$$\Delta\phi_{\mathrm{PPC}} = 42.981(1.0004118099),$$

which gives approximately

$$42.9987''/\text{century}\ .$$

Therefore, the second calculation is also internally consistent.

---

6. Why are the two β_{PPC} values different?

The proposed PPC perihelion value is kept fixed:

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''.$$

Only the GR reference changes:

$$42.980'' \to 42.981''.$$

Since

$$\beta_{\mathrm{PPC}} = 3\left(\frac{\Delta\phi_{\mathrm{PPC}}}{\Delta\phi_{\mathrm{GR}}} - 1\right),$$

changing $\Delta\phi_{\mathrm{GR}}$ necessarily changes β_{PPC}.

Consequently, the paper should not state simply that

$$\beta_{\mathrm{PPC}} = 1.305 \times 10^{-3}$$

without specifying the reference value.

The scientifically precise statement is:

> **Using the rounded GR reference value** $42.98''/\mathrm{century}$**, the proposed PPC perihelion value** $42.9987''/\mathrm{century}$ **corresponds to** $\beta_{\mathrm{PPC}} = 1.3052583 \times 10^{-3}$**. Using** $42.981''/\mathrm{century}$ **instead gives** $\beta_{\mathrm{PPC}} = 1.2354296 \times 10^{-3}$**.**

7. Scientific status of the calculation

There is an important distinction between **calculating β_{PPC} from a proposed perihelion value** and **deriving β_{PPC} from the PPC theory itself**.

The calculation above establishes:

$$\Delta\phi_{\text{PPC}} \longrightarrow \beta_{\text{PPC}}$$

once the GR reference value and the proposed PPC perihelion value are specified.

However, a fundamental PPC prediction would require the reverse theoretical chain:

$$E_d,\ P_g,\ \nabla P_g$$

$$\downarrow$$

$$g_{\mu\nu}^{\mathrm{PPC}}$$

$$\downarrow$$

$$\Gamma^{\alpha}_{\mu\nu}$$

$$\downarrow$$

Mercury's geodesic

$$\downarrow$$

$$\beta_{\mathrm{PPC}}$$

$$\downarrow$$

$$\Delta\phi_{\mathrm{PPC}}.$$

Using the rounded GR reference value,

$$\Delta\phi_{\mathrm{GR}} = 42.98''/\text{century},$$

and the proposed PPC value,

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''/\text{century},$$

the correction parameter is

$$\beta_{\mathrm{PPC}} = 1.3052583 \times 10^{-3}$$

or, rounded,

$$\beta_{\mathrm{PPC}} \approx 1.305 \times 10^{-3}\ .$$

However, using the more precise GR reference value,

$$\Delta\phi_{\mathrm{GR}} = 42.981''/\text{century},$$

with the same proposed PPC value,

$$\Delta\phi_{\mathrm{GR}} = 42.981''/\mathrm{century},$$

with the same proposed PPC value,

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''/\mathrm{century},$$

gives

$$\beta_{\mathrm{PPC}} = 1.2354296 \times 10^{-3}$$

or

$$\beta_{\mathrm{PPC}} \approx 1.23543 \times 10^{-3}\ .$$

Therefore, both numerical results are arithmetically correct, but they correspond to **different choices of the GR reference value**.

If the paper adopts the rounded GR benchmark of $42.98''/\mathrm{century}$, the principal result should be reported as:

$$\Delta\phi_{\mathrm{PPC}} = 42.9987''/\mathrm{century}$$

with

$$\beta_{\mathrm{PPC}} = 1.3052583 \times 10^{-3} \approx 1.305 \times 10^{-3}.$$

If instead the paper adopts $42.981''/\mathrm{century}$ as the primary GR reference, then the corresponding value should be reported as

$$\beta_{\mathrm{PPC}} = 1.2354296 \times 10^{-3} \approx 1.23543 \times 10^{-3}$$

2. Small pressure does not necessarily mean small pressure gradient

This is an important mathematical point.

A pressure field can have a very small value while changing extremely rapidly in space.

For example,

$$P_g(x) = \epsilon \sin\left(\frac{x}{\epsilon}\right)$$

where ϵ is very small.

The magnitude of pressure is bounded by

$$|P_g| \leq \epsilon,$$

so the pressure itself is very small.

But

$$\frac{dP_g}{dx} = \cos\left(\frac{x}{\epsilon}\right),$$

which can have magnitude as large as 1, **independent of how small ϵ becomes**.

Therefore,

$$\boxed{|P_g| \text{ small } \not\Rightarrow |\nabla P_g| \text{ small}}$$

and consequently,

$$\boxed{|P_g| \text{ small } \not\Rightarrow |\mathbf{F}_{\text{field}}| \text{ small}}$$

when

$$\mathbf{F}_{\text{field}} = -\nabla P_g.$$

Starting with

$$P_g = \omega E_d,$$

we have

$$\boxed{\nabla P_g = E_d \nabla \omega + \omega \nabla E_d}$$

and therefore

$$\boxed{\mathbf{F}_{\text{field}} = -E_d \nabla \omega - \omega \nabla E_d}.$$

This shows something particularly important: **the force-density field depends on spatial variation of ω and E_d, not simply on the magnitude of P_g.**

So even if P_g is locally very small, a sufficiently steep spatial variation can produce a substantial ∇P_g.

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This shows something particularly important: **the force-density field depends on spatial variation of ω and E_d, not simply on the magnitude of P_g.**

So even if P_g is locally very small, a sufficiently steep spatial variation can produce a substantial ∇P_g.

The **pressure-gradient field** is obtained by taking the spatial gradient:

$$\boxed{\nabla P_g = \nabla(\omega E_d)}$$

Using the product rule,

$$\boxed{\nabla P_g = E_d \nabla \omega + \omega \nabla E_d}$$

or explicitly,

$$\nabla P_g = \left(\frac{\partial P_g}{\partial x}\right)\hat{i} + \left(\frac{\partial P_g}{\partial y}\right)\hat{j} + \left(\frac{\partial P_g}{\partial z}\right)\hat{k}.$$

Thus,

$$\boxed{\nabla P_g = E_d \nabla \omega + \omega \nabla E_d}$$

Important case: $P_g = 0$

If

$$\omega E_d = 0$$

everywhere in the region, then P_g is identically zero and therefore

$$\boxed{\nabla P_g = \mathbf{0}}.$$

But if $P_g = 0$ **only at one point**, ∇P_g need not be zero.

Also, the ct dependence does **not** enter ∇P_g directly, because ∇ is the **spatial** gradient. Time variation would instead involve

$$\frac{\partial P_g}{\partial (ct)} = E_d \frac{\partial \omega}{\partial (ct)} + \omega \frac{\partial E_d}{\partial (ct)}.$$

F_field = F is Gravitational Pressure Field Force density.